

Faculty of Engineering, School of Electrical Electronics and Communication Engineering
 Department of Electrical Engineering
 Degree: B. Tech. Electrical & Computer Engineering Total Credit: 160

	Third Semester						Fourth Semester					
Code	Subject Name	L	T	P	C	Code	Subject Name	L	T	P	C	
MAS21XX	Statistics & Probability	3	0	0	3	MEE22XX	Engineering Economics	3	0	0	3	
MBB21XX	Management of Technology	3	0	0	3	ELC2201	Networks & Systems	3	1	0	4	
ELC2101	Analog & Digital Systems	3	1	0	4	ELC2202	Data Base Management Systems	3	1	0	4	
ELC2102	Data Structures & Algorithms	3	1	0	4	ELC2220/ ELC2221	Object Oriented Programming/ Microcontroller based Systems Design	3	1	0	4	
ELC2103	Computer Organization & Architecture	3	1	0	4	ELC22XX	Program Elective 1	3	0	0	3	
ELC2120/ ELC2121	Electrical Vehicle Technology/ Computer Networks	3	1	0	4	ELC00XX	Open Elective 1	3	0	0	3	
ELC2130	Data Structures & Algorithms Lab	0	0	2	1	ELC2230	Data Base Management Systems Lab	0	0	2	1	
ELC2131	Analog & Digital Systems Lab	0	0	2	1	ELC2231	Microcontroller Lab	0	0	2	1	
ELC2170	Project-based Learning 1	0	0	2	1	ELC2270	Project-based Learning 2	0	0	2	1	
		18	4	6	25			18	3	6	24	
	Total Contact Hours (L+T+P)	28					Total Contact Hours (L+T+P)	27				
	Fifth Semester						Sixth Semester					
Code	Subject Name	L	T	P	C	Code	Subject Name	L	T	P	C	
ELC3101	Power Electronics	3	1	0	4	EEE3201	Control Systems	3	1	0	4	
ELC3102	Smart Energy Systems	3	1	0	4	ELC32XX	Program Elective 4	3	0	0	3	
ELC3120/ ELC3121	Renewable Energy / Operating Systems	3	1	0	4	ELC32XX	Program Elective 5	3	0	0	3	
ELC31XX	Program Elective 2	3	0	0	3	ELC32XX	Program Elective 6	3	0	0	3	
ELC31XX	Program Elective 3	3	0	0	3	ELC00XX	Open Elective 3	3	0	0	3	
ELC00XX	Open Elective 2	3	0	0	3	ELC3230	Professional Practice	0	0	2	1	
ELC3130	Power Electronics Lab	0	0	2	1	ELC3231	Control & Automation Lab	0	0	2	1	
ELC3131	Energy Systems Lab	0	0	2	1	ELC3232	Advance Systems Simulation Lab	0	0	2	1	
ELC3170	Project-based Learning 3	0	0	2	1	ELC3270	Project-based Learning 4	0	0	6	3	
		18	3	6	24			15	1	12	22	
	Total Contact Hours (L+T+P)	27					Total Contact Hours (L+T+P)	28				

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Seventh Semester						Eighth Semester					
Code	Subject Name	L	T	P	C	Code	Subject Name	L	T	P	C
ELC41XX	Program Elective 7	3	0	0	3	ELC4270	Major Project	0	0	24	12
ELC41XX	Program Elective 8	3	0	0	3						
ELC00XX	Open Elective 4	3	0	0	3						
ELC00XX	Open Elective 5	3	0	0	3						
ELC4170	Internship (Industry or Research)	0	0	2	1						
		12	0	2	13			0	0	24	12
	Total Contact Hours (L+T+P)	14					Total Contact Hours (L+T+P)	24			

Flexi Core		
Flexi Core 1 (III Sem)	Flexi Core 2 (IV Sem)	Flexi Core 3 (V Sem)
ELC2120: Electrical Vehicle Technology ELC2121: Computer Networks	ELC2220: Object Oriented Programming ELC2221: Microcontroller based Systems Design	ELC3120: Renewable Energy ELC3121: Operating Systems

Program Electives			
IV Sem	V Sem	VI Sem	VII Sem
PE1 • ELC2240: Solar Photovoltaic systems • ELC2241: Generation, Transmission & Distribution • ELC2242: Graph Theory & Applications	PE2 • ELC3140: Engineering Systems Modelling • ELC3141: Soft Computing Techniques • ELC3142: Internet of Things PE3 • ELC3143: Software Engineering • ELC3144: Fuzzy Logic and Neural Network • ELC3145: Digital Signal Processing	PE 4 • ELC3240: Data Analytics • ELC3241: Sensor & Actuator • ELC3242: Fundamentals of Semiconductor Devices PE5 • ELC3243: AI and Machine Learning • ELC3244: Introduction to Blockchain • ELC3245: Industrial Automation PE6 • ELC3246: Modern Optimization Techniques	PE 7 • ELC4140: Cloud Computing • ELC4141: Energy Markets & Operations • ELC4142: VLSI Design & Applications PE8 • ELC4143: Web Technology • ELC4144: Cyber Physical Systems • ELC4145: Energy Audit and Management

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		• ELC3247: Microgrid Technology • ELC3248: Forecasting Methods and Applications	
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Open Electives	
Graded OE	Non-Graded OE
OE1 ELC0001: Fundamentals of Electric Vehicle OE2 ELC0002: Fundamentals of Solar PV Systems OE3 ELC0003: Battery Management Systems OE4 ELC0004: Renewable Energy Systems OE5 ELC0005: Energy Auditing & Management	OE1 ELC0051: Course Name OE2 ELC0052: Course Name OE3 ELC0052: Course Name OE4 ELC0052: Course Name OE5 ELC0052: Course Name